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Introduction

Why do news stories feel different in each newspaper?

- One newspaper might call a protest 'জনগণের ন্যায্য দাবি' while another calls it “বিশৃঙ্খলা”
- Same event, completely different narrative.



Why BiasFree News?

- Many Bengali news headlines are **sensational or politically influenced**
- Most readers don't even realize they are **consuming biased content**
- **Clickbait headlines** don't always match the real article
- **No strong tool exists** for detecting bias in Bangla news
- Manual **fact-checking can't keep up** with daily article volume
- Most readers are no longer **able to distinguish fact from opinion**

What is BiasFree News?

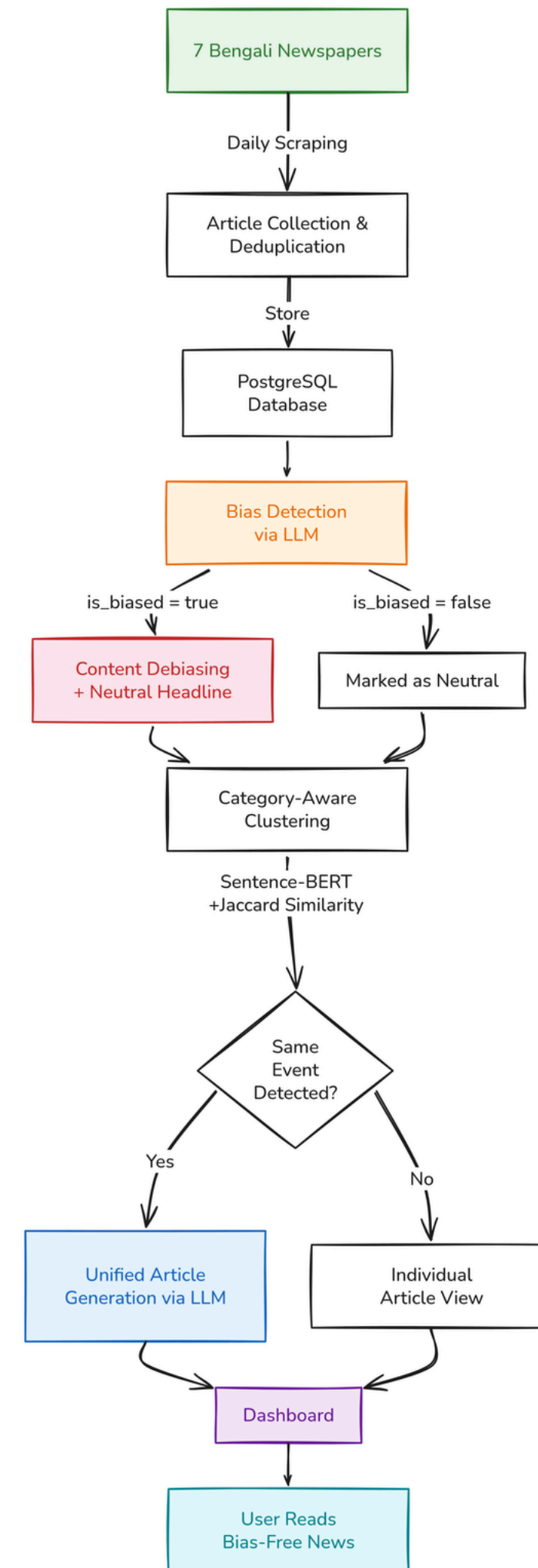
A **web** platform that:

- Scrapes articles daily from **7 Bengali newspapers**
- Detects bias using **LLM** – scores each article 0–100
- **Neutralizes biased content** while preserving meaning
- Clusters articles from **different sources covering the same event**
- Generates a **single unified**, bias-free article per event

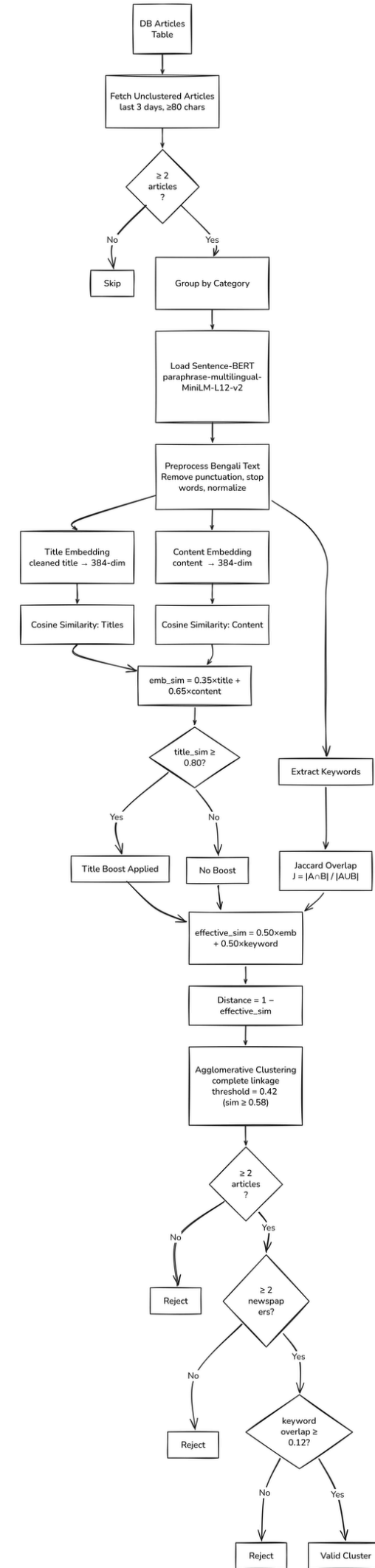
Quality Function Deployment

Normal Requirements	Expected Requirements	Exciting Requirements
• Bias detection in Bengali news using LLM	• Side-by-side original vs. debiased text with highlights	• Similar article detection using semantic similarity
• Debias content by replacing biased terms with neutral alternatives	• Historical archive with filtering by source, date, and bias level	• Unified neutral article generation from multiple sources
• Generate neutral, content-aligned headlines	• Detailed explanations for why each term was flagged	• Interactive dashboard with bias trends, source statistics, real-time monitoring
• Scrape articles from 7 major Bengali newspapers automatically	• User authentication with email verification & password reset	
• News classification into categories	• Automated scheduled scraping (daily)	
• User analysis history with filtering & sorting	• Priority-based category sorting (personalized feed)	6

Workflow



Applied Metholodology



Project Demonstration

Test case Check

Future Work

Future Work

- **Multi-language support** – extend beyond Bengali (English)
- **Real-time bias detection** – browser extension for instant analysis while reading
- **User feedback loop** – let readers flag incorrect bias detections to improve model accuracy
- **Source credibility scoring** – long-term tracking of bias patterns per newspaper

Thank You!

Challenges

Challenges

- Bengali NLP complexity – embeddings show high false-positive similarity (~ 0.97) for unrelated articles in the same category; solved with Jaccard keyword overlap as a lexical anchor